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Optimization of multimodal networks including environmental costs: A model and findings for transport policy

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ABSTRACT

In this paper we introduce a modelling approach for the optimization of terminal networks, taking into account the costs of CO_2 emissions and economies of terminal scale. The optimization problem is solved by bi-level programming, where the upper level searches for the optimal terminal network configurations by using a genetic algorithm, while the lower level performs multi-commodity flow assignment over a multimodal network.

This model is applied to optimize the Dutch container terminal network configuration. The model was calibrated using both the national survey data of freight transport flows by mode and data on throughputs of the Dutch inland container terminals. We find that different efficient terminal network configurations result under different pricing regimes. Also, a reduction in total system costs appears feasible when CO_2 emission prices are raised, by taking advantage of previously unexploited economies of scale in the system.

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1. Introduction

The pre-crisis growth of intercontinental freight flows has resulted in major investments in port and hinterland capacity. In many ports in the world, road systems are saturated. Nowadays port development relies increasingly on railway and waterway systems for hinterland access. Intermodal transhipment is needed to allow access to, and egress from regions that are not directly accessible by rail and waterways. In the current time of the growth of maritime transport, keeping up with changing demand is an important task for the freight transport industry. Intermodal transport networks have taken advantage of the recent strong growth of maritime freight transport. Obviously, the demand for transport capacity depends not only on the growth of the world economy. Terminal networks also have to adapt to changes in flow patterns over time in terms of geography and commodity type. Technological innovations in handling equipment and information technology may result in new network services, and translate into changes in intermodal service quality.

In our paper we describe a new model for the design of intermodal freight transport networks. Our aim is to understand how terminal networks should be designed to accommodate a specific container transport demand, specified by total volume and commodity composition. In the design we take into account infrastructure investments (terminals and links) and transport pricing policies. Concerning the latter, our hypothesis is that differences between modes in terms of CO_2 emissions can imply different uses of terminals, depending on their location in the multimodal network. Hence, CO_2 emission prices may also affect the optimal terminal network configuration. The model presented here enables this effect to be studied. To test its validity and demonstrate its use for applications, the model was implemented for the case of the Dutch container terminal network. This large and dense network mainly serves the hinterland container transport flows of the Port of Rotterdam.

The paper is structured as follows. We present the model in Section 2 of the paper, including its modules representing the demand and supply side of the market, the cost functions, the GIS environment, and the optimization process. We apply the model in Section 3 to the case of Dutch container terminal network design. We summarize our findings and conclude the paper in Section 4.

2. The multimodal network design model

The objective of multimodal network design is to identify the network structure (number and location of terminals) that

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minimizes network costs, given transport demand. Usually overall network demand is assumed to be fixed. Network costs generally include the out-of-pocket transport costs (or transport prices to the shipper), without inventory costs or external costs. Further influences on terminal network structure may include transport policies concerning local constraints on development or the application of fiscal instruments, e.g. investment subsidies or transport pricing. Intermodal freight networks have been studied in the past, although reports in the literature of design methods geared towards real-world application are rare (for reviews see [1–3]). Recent studies of network development can be found in the area of: service network design [4]; terminal network design from a private perspective [5]; and public intermodal networks [6,7]. Problematic issues identified in these applied network design modelling studies included:

- scale economies, resulting in multiple locally optimal networks;
- linkages between service networks and infrastructure networks;
- lack of a validated demand model;
- long computation times in real world applications.

Heuristic solutions often constrain the flexibility in the optimization model for the types of commodity or service networks that can be taken into account, or do not allow solutions to be simultaneously identified for the number and the location of terminals. Solution techniques such as simulated annealing and genetic algorithms (GAs) allow some flexibility in problem formulation but have drawbacks in terms of tractability of solutions or computation times. The newly developed method presented in our paper uses a GA-based approach to solve a comprehensive network design problem, which takes into account economies of terminal scale, multiple commodities, and multiple modes.

The model consists of four modules (see Fig. 1): (1) the demand module with origin and destination (OD) matrices embedded; (2) the supply module with European geographic information and the network features database; (3) the optimization module which optimizes the terminal network in terms of terminal number, capacity, and location, with the objectives of minimizing the system generalized costs and minimizing the system CO_2 emissions; and (4) the policy package generator which generates policy packages. Each policy package consists of a terminal network configuration and a CO_2 emission price. Each policy package is one potential solution for the optimization module.

2.1. Demand module

The demand module consists of databases of OD transport demands per mode per commodity group on an annual basis. The commodity groups are defined on the basis of the value of the commodities. This will be further discussed in Section 2.3. The OD regions are represented in GIS via the geometric centroids of the regions. The relation between the geometric centroid of a region and the centroid of the transport demand of the region will be elaborated via the definition of 'access/egress' (see Section 2.2) and its cost function (see Section 2.3).

The regional demands are modelled based on the topology published by Eurostat [8]. The centroids of the regions at levels NUTS 0, 1, 2, and 3 (the Nomenclature of Territorial Units for Statistics levels) [9] are defined in GIS and connected to the transport networks. This way of OD representation provides flexibility for data fusion when multiple data sources with different statistics levels have to be applied. In addition, it gives the possibility of outputting aggregated results at various levels for various regions.

2.2. Supply module

The supply side of transport is described in terms of capacity, accessibility, and price in this model. The capacity and the accessibility are identified by: the physical capacity of the infrastructures; the geographic accessibility of the infrastructures; and the frequencies of service provided by the transport operators. Since this model is designed for long-term terminal network design the focus is put on infrastructural supply modelling. Scheduling and fleet arrangement are not taken into account.

2.2.1. GIS modelling of the infrastructure supply

A GIS-based European multimodal freight transport network is developed in the TransCAD® environment by using 'virtual network' modelling [10–13]. The infrastructure networks of three transport modes (road, rail, IWW) are connected via terminals. The intermodal accessibility at each terminal is specified according to the transhipment services provided by the terminal. Fig. 2 shows the components of the virtual multimodal network.

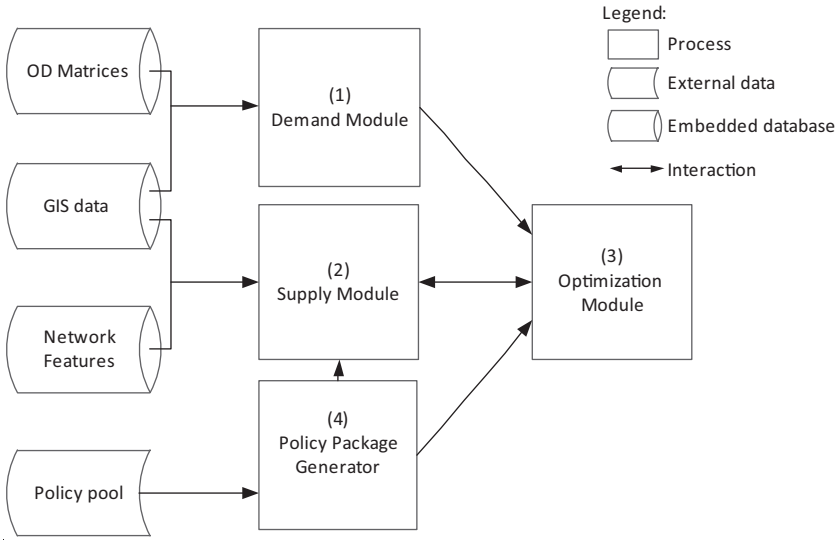


Fig. 1. General framework of the model.

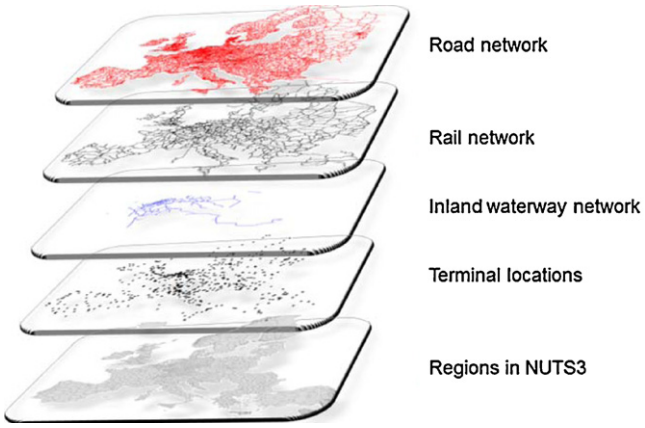


Fig. 2. Visualization of the European road network, rail network, IWW networks, terminal locations, and OD regions at the NUTS 3 level. *Notes*: GIS data of road, rail, IWW network sourced from the GIS database of TransTools [14].

The virtual network can be described as $Z = (N, L)$ and L denote, respectively, the identifications (IDs) of the nodes of the network and the links of the network. Each link is associated with its fore-node and end-node. Each node is identified by its geographic coordinates. The nodes representing the ends of the real geographic links (i.e. a segment of road, rail, or IWW) are denoted as $n_i^m (i \in N, m \in M)$ · m denotes the mode of the geographic link to which the node belongs, $M = \{road, railway, IWW\}$. The nodes representing the terminals are denoted as $n_i^t (t \in N,\ t \in T)$. A terminal is tagged by a terminal ID t. Set T contains the IDs of all terminals. The accessible modes of the terminal are attached to $t \in T$. Besides the fore-/end-nodes of the real geographic links and the terminals, a centroid $n_i^o (i \in N,\ o \in O)$ is identified by the geographic coordinates of the geographic geometric centroid of the region. The centroids are virtual nodes in the network. They do not represent any infrastructure but each centroid represents the existence of the potential transport demand of a region, and thus each centroid is associated with the ID of a region, $o \in O$. O consists of the set of the IDs of all regions. The virtual network is constructed by connecting the nodes according to certain rules. Using the categorization of the nodes, links can be categorized automatically based on the node types of their fore-nodes and end-nodes. In this module, four sets of links are generated, $L = XUHUGUE$. The link $l_x^m (x \in X \subset L, m \in M)$ represents the real geographic link x of mode m. The transhipment links $l_h^{t,m(t)} (h \in H \subset L, t \in T, m \in M)$ are generated by connecting the terminal node n^t to a number of the closest nodes of mode m. m represents the modes which are accessible at the terminal t. The links that describe the pre-/end-haulages between terminal t and centroid o are denoted as $l_g^{ot} (g \in G \subset L, t \in T, o \in O)$. They are generated by connecting the centroid n^o to the terminals located within a certain geographic radius of centroid o. The radius is region-specific and is determined by the maximum pre-end/haulage of the shipments starting or ending in this region. The centroids are also connected to the terminals within a predefined radius by virtual links (centroid-terminal connector). The access/egress links $l_e^o (e \in E \subset L, o \in O)$ describe the access/egress between the road network and region o. The length of an access/egress link indicates the average distance of accessing from any place of the region to the road network and egressing from an appropriate segment of the road network to a destination in the region. Fig. 3 is a graphical representation of the network.

The entire virtual network (see Fig. 4) is composed of 32,747 nodes (including 31,368 geographic nodes, 659 terminals, and 720 centroids) and 41,522 links (including 40,041 geographic links, 1365 transhipment links, 762 pre-/end-haulage links, and 720 access/egress links). The geographic data, e.g. the geographic coordinates of the geographic nodes, the length and mode of the geographic links, and the topology were adapted from the GIS database of TransTools [14]. 659 terminals providing intermodal transhipment services are connected to the virtual network. The geographic coordinates of the terminals were obtained by geocoding the addresses of the terminals.

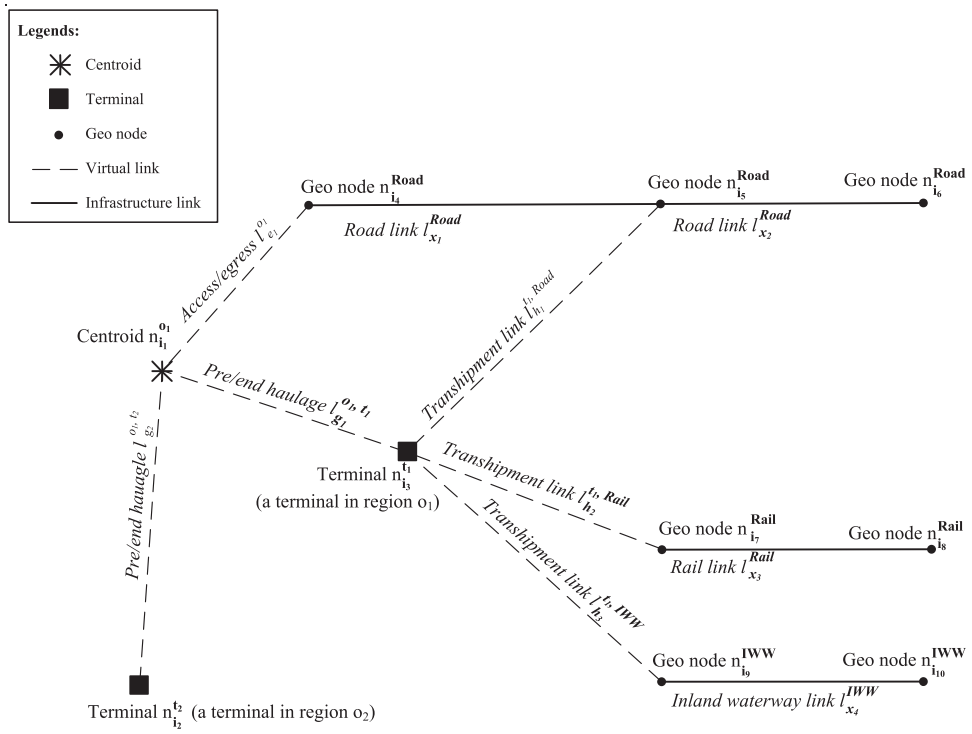


Fig. 3. Outline of network representation.

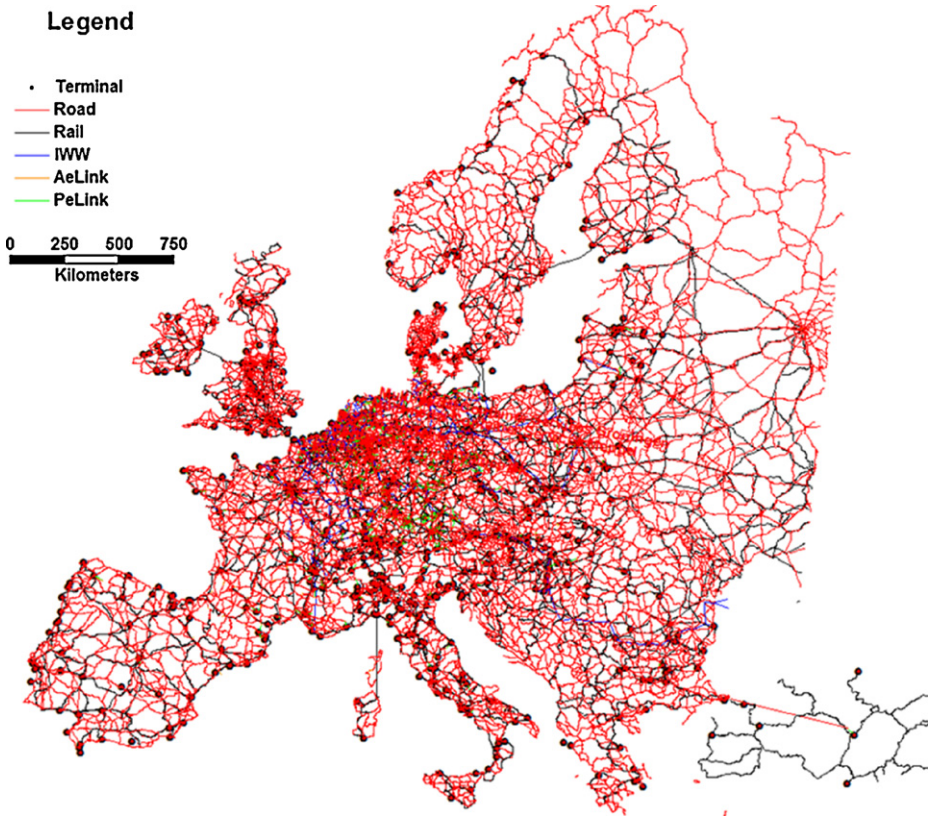


Fig. 4. The GIS-based multimodal European transport network. *Notes*: GIS data of road, rail, IWW network sourced from the GIS database of TransTools [14].

The generalized costs for moving one unit of each type of commodity are calculated based on the cost structures in practice for each transport mode and transhipment, to which with the costs of CO_2 emissions added. Four different cost functions are applied to calculate the generalized costs of the four types of link. The results are attached to the corresponding links in the virtual network for calculating the generalized route costs.

2.2.2. Generalized costs of geographic links

The generalized costs of moving one unit of commodity p over geographic link x is formulated as:

$$A^{px} = \kappa^{am} \cdot s_x + \tau^p \cdot \left(\frac{s_x}{\upsilon^{am}}\right) + \lambda^m \cdot s_x \cdot c_{CO_2}, \quad \forall x \in X \subset L, \tag{1}$$

where κ^{am} is the unit mode-related costs of mode m in main haulage; τ^p is the unit freight-related time costs of commodity p; s_x is the length of link x; υ^{am} is the average speed of mode m in main haulage; λ^m is the tonne-km CO_2 emissions by mode m and c_{CO_2} is the price of CO_2 emissions per tonne.

Mode-related costs κ^{am}, commodity-related costs τ^p, and CO_2 emission costs c_{CO_2} are the components of the generalized costs of geographic links. The mode-related costs are further specified in two parts: (1) the distance-related part, which depends on the distance between the origin and the destination, including maintenance, repairs and fuel; and (2) the time-related part, which is the time costs depending on the transport mode, directly borne by the transport operators, including depreciation, interest, facility tax, licences and permits, insurance, wages, and business costs. The commodity-related time costs mainly include interest, depreciation, and loss of the market value of the freight during the transport process. They are independent of the transport mode, and lead to heterogeneous mode choices when moving different commodities between the same OD pair. The transport operators of the commodities with high commodity-related time costs have a high likelihood of choosing a fast transport mode. Therefore, we used commodity-related time costs to describe the heterogeneity of multicommodity in mode choice. We categorized the total volume of the containers into 5 commodity groups based on their commodity-related time costs.

2.2.3. Generalized costs of transhipment links

The generalized costs of moving one unit of commodity p over transhipment link h is formulated as

$$B^{ph} = f(\omega_h, \nu_h) + \lambda^h \cdot c_{CO_2} + \varepsilon_h^t, \quad \forall h \in H \subset L, \tag{2}$$

where ω_h is the unit handling cost of transhipment link h; ν_h is the flow over transhipment link h; λ^h is the CO_2 emissions incurred in transhipment h and ε_h^t is a parameter indicating the various costs caused by specific features of the handling between terminal t and a certain transport mode via transhipment link h.

Transhipment costs are not defined by time-related costs or distance-related costs, owing to the complexity of terminal activities. Depending on the services provided by a certain terminal, there might be other value-added services provided during the period of transhipment (e.g. storage). The cost structure of transhipment is different from transport costs. Despite the emission costs, the transhipment costs borne by the transport operators are defined as the costs borne by the terminal operators with the assumption of complete competition and long-term operation. The costs of terminal operators are the handling costs which depend on the terminal throughput and the scale of the terminal. For a terminal operating under its capacity, an increase in throughput results in reduced unit handling costs, known as the "economies of density". If throughput continuously increases and causes congestion in the terminal, the inefficiency in handling would result in an increase in unit handling costs. In order to avoid this inefficiency, the terminal could invest in capacity enlargement

and technology improvement. So a larger terminal with high throughput is more likely to achieve lower unit handling costs compared with a smaller scaled terminal. The cost efficiencies gained by scale are known as "economies of scale". Principally, in the long run, the economies of density, the inefficiency caused by capacity shortage, the scale enlargement and handling technology improvement, the diseconomies of density after scale enlargement, and the economies of scale appear iteratively. Therefore, the unit handling costs decrease when the throughput (flow over the terminal) increases in general but not monotonically. Ballis and Golias [15] show the relationship between terminal scale and handling cost by a group of convex curves. In order to simplify the computation induced by the flow-related cost function, we ignored the diseconomies of scale associated with capacity enlargements and applied a monotonic decreasing function to describe the handling costs dynamics. Because terminal operation and administration efficiencies are not the same at each terminal, the transhipment costs of different terminals may differ, even if they have the same scale and have the same throughput. These features of a terminal are modelled by ε_h^t in the function Eq. (2).

2.2.4. Generalized costs of access/egress links and pre-/end-haulage links

The generalized costs of moving one unit of commodity p over access/egress link e is formulated as:

$$D^{pe} = \kappa^{\gamma} \cdot s_e + \tau^{p} \cdot \left(\frac{s_e}{\upsilon^{\gamma}}\right) + \lambda^{\gamma} \cdot s_e \cdot c_{CO_2} + \varepsilon^{o}, \quad \forall e \in E \subset L, \tag{3}$$

where κ^{γ} is the unit mode-related cost of access/egress; s_e is the distance of access/egress link e; υ^{γ} is an average speed of access/egress; λ^{γ} is the CO_2 emissions per tonne-km of access/egress and ε^{o} is a parameter indicating the various costs caused by specific features of the access/egress between region o and the road network.

The generalized costs of moving one unit of commodity p over pre/end-haulage link g is formulated as:

$$C^{pg} = \kappa^{\beta} \cdot s_g + \tau^{p} \cdot \left(\frac{s_g}{\upsilon^{\beta}}\right) + \lambda^{\beta} \cdot s_e \cdot c_{CO_2} + \varepsilon^{ot}, \quad \forall g \in G \subset L, \tag{4}$$

where κ^{β} is the unit mode-related cost of pre-/end-haulage; s_g is the distance of pre-/end-haulage g; υ^{β} is an average speed of pre-/end-haulage; λ^{β} is the CO_2 emissions per tonne-km of pre-/end-haulage and ε^{ot} is a parameter indicating various costs caused by specific features of the pre-/end-haulage between region o and terminal t.

Access/egress usually means the trucking haulage between an origin/destination and one of the highway, rail, or IWW networks [12,16,17]. The term 'access/egress' is exclusively used to describe the highway access in this model because the costs derive from different transport activities, and are influenced by different factors when accessing/egressing to/from the highway network or a terminal. The access/egress costs between the highway network and the region centroid describe the average transport costs of trucking between any location within the region and a highway entrance. There are region-related costs which depend on the area of the region, level of road facilities, average traffic density, and other geographic features (e.g. mountain area or delta area). In addition, we used the term 'pre-/end-haulage' to describe the haulages between a location within the region and a terminal. They are terminal-related and region-related costs. In addition to the influencing factors of the highway access/egress costs, the pre/end-haulage costs also depend on the relationship between the region and the terminal, e.g. the density of the transport demand generated close to the terminal. We modelled the varieties of the access/egress costs per region and the pre/end-haulage costs per region-terminal pair by the stochastic parameters ε^{o} in function Eq. (3), and ε^{ot} in function Eq. (4).

2.3. Network optimization module

The main function of this model is to find the optimal terminal network configurations under different CO_2 emission prices with the objectives of minimizing system costs and minimizing system emissions. Here, the system costs mean the total generalized costs with the internalized CO_2 emissions charges of all the flows accommodated by the virtual network. A terminal network configuration includes the total number of terminals and the location of each terminal with its associated capacity. Each terminal network configuration presents a specific scenario of terminal network design. Internalizing the CO_2 emissions costs by introducing CO_2 emission prices may lead to modal shift because transporting the same volume of freight via rail or inland waterway over the same distance generates less CO_2 emissions compared with road transport. The more expensive the CO_2 emissions, the more preferable are rail and/or inland waterway transport, and thus the less the system emissions. Meanwhile, the generalized system costs with the costs of CO_2 emissions internalized increase along with the increase in the CO_2 emission price. So there is a trade-off between minimal system costs and minimal system CO_2 emissions. In addition, assigning the given and fixed OD demands over various candidate terminal network configurations may lead to different uses of the network. For different CO_2 emission prices there are different optimal terminal network configurations. Therefore, we argue that there is a frontier of equilibriums of minimum system costs and minimum system CO_2 emissions under various terminal network configurations. The frontier is expected to be found by evaluating and comparing the system costs and system CO_2 emissions of the candidate terminal network configurations associated with various CO_2 emission prices.

This network optimization problem is modelled by using bi-level programming. The lower level models the multi-commodity flow assignment over a multimodal container transport network taking economies of terminal scale into account. The flow assignment is constrained by the terminal accessibilities defined at the upper level, while the upper level searches for optimal policy packages which leads to minimal system costs and minimal CO_2 emissions at the lower level. The bi-level programming is formulated as follows:

Upper level:

$$MIN(\theta_1 \cdot \text{system costs} + \theta_2 \cdot \text{system } CO_2 \text{ emissons}) = f(a(\delta, b), c^*, v^*, c_{CO_2}) \tag{5}$$

where a is the two-dimension array representing terminal network configurations; δ is the binary array expressing the openness of the terminals, $\delta_{ki} = 1$ if terminal i is opened in terminal network configuration a_k, where $i \in n$, the total number of candidate terminals, and $k \in m$, $m = 2^n$; $\boldsymbol{b}$ is an array of terminals capacities. b_{ki} is the capacity of terminal i in terminal network configuration a_k; $\boldsymbol{c}^*$ is the vector of generalized link costs of the optimal solution at the lower level; $\boldsymbol{v}^*$ is the vector of link flows of the optimal solution at the lower level; c_{CO_2}: CO_2 emission price; θ_1 and θ_2: the weights of system costs and CO_2 emissions, respectively.

Lower level:

$$MIN \text{ total cost} = \sum_p \sum_l (c^{pl} \cdot v^{pl}), \tag{6}$$

where c^{pl} is the generalized costs of link l when moving one unit of commodity p and v^{pl} is the flow of commodity p over link l.

The flow assignment model applied at the lower level is a simple form of multimodal freight transport network planning models [1,18]. Observing that the freight generated in one region can be transported via the terminals located within the region and/or the terminals located in the neighbouring regions, we modelled terminal choice in addition to mode choice and route choice. Over the virtual multimodal network, the combined mode-terminal-route choice is carried out in one procedure based on the rule of generalized route costs minimization.

The generalized route costs of moving commodity p over route r is formulized as follows:

$$GCR_r^p = \left(\sum_{x \in R \cap X} A^{px} + \sum_{h \in R \cap H} B^{ph} + \sum_{g \in R \cap G} C^{pg} + \sum_{e \in R \cap E} D^{pe} \right) \cdot v_r^p, \quad \forall X, H, G, E \subset L. \tag{7}$$

A route is composed of the link set R. Based on the representation of the network, each link belongs to one of the four sets of links, i.e. geographic links X, transhipment links H, pre-/end-haulage links G, and access/egress links E. The generalized costs of moving one unit of commodity p over link l is described by A, B, C, or D, depending on to which link set l belongs. The aggregation of the unit generalized costs incurred on the links belonging to link set R is the generalized costs of moving one unit of p over route r. v_r^p denotes the flow of p over route r. It should be noted that we assumed the transport operators as the end-users of the container transport network, and the relevant costs borne by the transport operator are used to describe the price of the transport service supply.

In order to test the impact of passenger flows on freight flow assignment, an experiment has been carried out on freight flow assignment with preloaded passenger flows for areas where congestion is regularly observed. The result shows no capacity shortage on an annual basis. Since the model is designed for freight flow assignment with the fixed demand given on an annual basis, we assumed that passenger flows have no impact on freight flow assignment. The impacts of congested areas are captured by lower average transport speed of the recognized congested road segments.

An all-or-nothing (AON) algorithm is used in this model for the flow assignment. Principally, an AON algorithm is used to assign the demand of each OD pair to the route between the OD pair and with the lowest generalized route costs. Two limits of the AON algorithm are that: (1) the capacities of the candidate routes are assumed to be unconstrained; and (2) the heterogeneity of the route choosers is not taken into account. Capacitated assignment algorithms and equilibrium algorithms with or without stochasticity are commonly used in freight transport network planning. However, here we chose AON algorithm for the assignment for three reasons:

1. Given the assumption of no capacity shortage, the results from the AON algorithm or user equilibrium are highly likely to be the same. Using the AON algorithm saves much computing time due to that it requires no iteration to achieve the equilibrium.
2. Stochastic parameters are defined for each type of link. No extra stochastic parameter needs to be introduced in the assignment process, therefore stochastic algorithms are not suitable.
3. Taking terminal scale economies into account, the cost function may be concave, and therefore the assignment may not able to achieve an equilibrium.

Therefore, a derivative AON algorithm based on the Dijkstra method [19] is used for route choice and flow assignment. The AON algorithm is provided by the TransCAD® software. A genetic algorithm (GA) [20] is used in solving the optimization problem at the upper level, because of the numerous number of candidates, and the non-convex cost function caused by economies of terminal scale. The GA used in this model is self-scripted in GISDK (the second development environment of TransCAD) on the basis of the programming logic developed by Michalewicz [21]. We designed our GA by applying 'roulette wheel selection', 'n-point half-uniform crossover', 'uniform mutation', and 'elitist' strategies [22–25]. The iterative procedure is described as follows.

Inner loop: flow assignment
Step 1: Initiate the network features according to the reference scenario.
Step 2: Load solution j, $j \in [1,$ *number of solutions of this generation*$]$ of generation i of the network. The transhipment costs of each terminal is calculated according to the predefined flow-related cost function.
Step 3: Run the multicommodity AON flow assignment.
Step 4: Update transhipment costs of each terminal based on the flow assigned to the terminal in this run and the terminal cost-flow function.
Step 5: Iterate step 3 and step 4 until the difference of the transhipment costs in two adjacent runs is less than the predefined tolerance at every terminal.
Step 6: Calculate the system costs and the CO_2 emissions of this solution.
Step 7: If $j + 1 =$ *the number of solutions of this generation*, then go to Step 2 of the outer loop.
Step 8: Load solution $j + 1$ of generation i to the network, and then go to Step 3.
Outer loop: optimization
Step 1: Generate the first generation of solutions. Go to inner loop for solution 1 of this generation.
Step 2: Find the "best" solution of this generation.
Step 3: If $i + 1 =$ *the number of generation*, then go to Step 6.
Step 4: Generate the next generation by applying GA techniques 'roulette wheel selection', 'half-uniform crossover', 'uniform mutation', and 'elitist'.
Step 5: Iterate Steps 2–5.
Step 6: Return the "best" of all solutions of all generations, and stop.

2.4. Policy package generator

The policy package generator generates the first generation of the candidate policy packages which are to be used as the starting points in GA in the optimization module. A policy package consists of a terminal network configuration and a CO_2 emission price which are, respectively, expressed by $a(\delta, b)$ and c_{CO_2} in Eq. (1). The total number of candidate terminals and their locations is defined manually. In other words, the number of elements of array δ and the correspondence between each element of δ and each candidate terminal are given and fixed. Each element of array a is one terminal network configuration. It defines which candidate terminals are to be opened in the current policy package, as well as the capacity of each opened terminal. The range and accuracy of CO_2 pricing policy are given via array c_{CO_2}.

3. Application: optimal terminal network configuration

The model is applied to optimize the Dutch container terminal network configuration. In 2006, there were approximately 50 Dutch terminals providing proper container handling services with throughputs ranging from 5000 TEUs to more than 500,000 TEUs. This section presents the frontier of terminal network configurations where equilibria can be achieved between minimum system

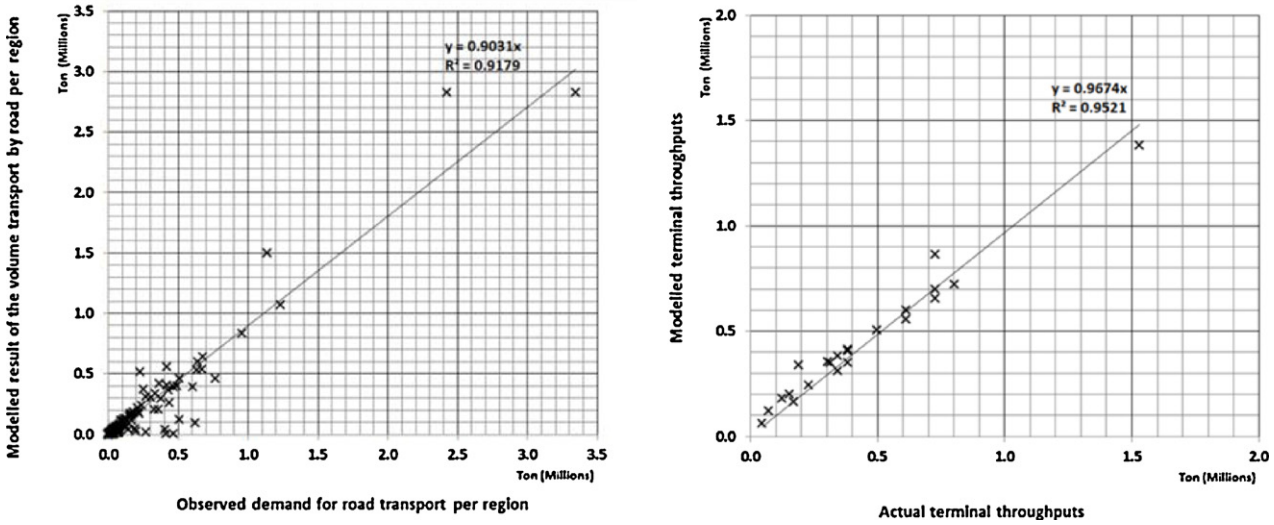


Fig. 5. Calibration results of mode choice and terminal choice.

generalized costs and minimum system CO_2 emissions. It also discusses the estimation of the trends of the system generalized cost change and the system CO_2 emission change if CO_2 emissions are priced at different rates (here, for the purpose of illustration, we vary prices in the range €10–€1000 per tonne). The scope of the system was defined as the Netherlands. Both domestic and intra-European flows were taken into account in the flow assignment; the evaluation of the generalized costs and emissions were, however, on the basis of the flows accommodated within the Dutch borders.

3.1. Demand and supply data

For our specific application, the database of container transport demand of the Netherlands in 2006 was loaded into the demand module of the model. This database was developed on the basis of a survey provided by the Central Statistics Bureau of the Netherlands [26]. The survey records approximately 24,000 container shipments transported to/from/through the Netherlands. The cost information required by the supply module of the model was collected from various sources. Previous research shows a wide range in mode-related transport and transhipment costs [27–30]. We first summarized the mode-related transport costs and transhipment costs with ranges from the literature, and then employed a GA-based searching process to determine the best combination of the cost values which leads to good fit of the modelled flow distribution to the observed flow distribution. Because no information about time value per commodity value group was available, we estimated the commodity-related time costs by redistributing the trade value per NSTR group [31] over transport volume [26], and relating the trade value to the commodity-related time cost via redistributing the commodity-related time cost per mode [32,33] over transport volume. The CO_2 emissions per tonne-km of road and rail transport were calculated based on the NEA report [34]. The tonne-km CO_2 emissions of barges in Class IV [35], and operating at a speed of 12 km/h in the waters with a draft of 2 m, is used as the transport emissions of inland waterways, which were calculated based on the Holtrop and Mennen method [36] with a correction for shallow-water effects. The CO_2 emissions in container transhipment at the inland terminals in the port area of Rotterdam are used for all transhipments in this model [37]. The handling costs of the terminals at various scales were estimated on the basis of the results of previous research [6,15,27,38,39] and the actual costs of the Dutch container terminals. A regression of unit transhipment costs on terminal throughputs for the Dutch container terminals was estimated and applied in this case study. The regression function is $y = 26.285\,x^{-0.146}$, where y denotes unit transhipment costs, and x denotes terminal throughput. The regression shows strong scale effects for terminals in the typical range of operation up to 3 million tonnes/year.

3.2. Model calibration and goodness-of-fit

The flow assignment model was calibrated by estimating the stochastic parameters in the cost functions of transhipment cost, access/egress cost, and pre-/end-haulage cost, i.e. vector ε^t, ε^o, and ε^{ot}. The calibration was done by employing the genetic algorithm as well. Fig. 5 shows the results of the calibration of the flow assignment model. The performance of the model on mode choice was evaluated by comparing the modelled modal split per OD pair with the modal split concluded from the freight transport survey 2006 [26]. The terminal choice was evaluated by comparing the modelled throughputs of the Dutch inland terminals with the actual throughputs of the terminals which were able to provide reliable data. The fit of the model was satisfactory, reaching high correlation coefficients between observed and calculated values for terminals and for transport links ($R^2 > 0.9$).

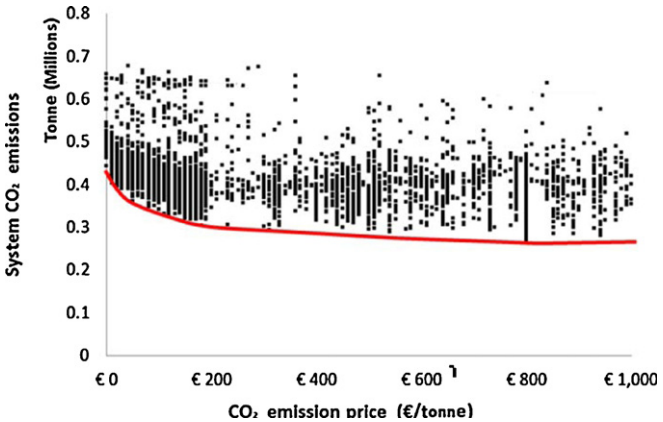


Fig. 6. System CO_2 emissions under CO_2 emission prices ranging €0–€1000 per tonne.

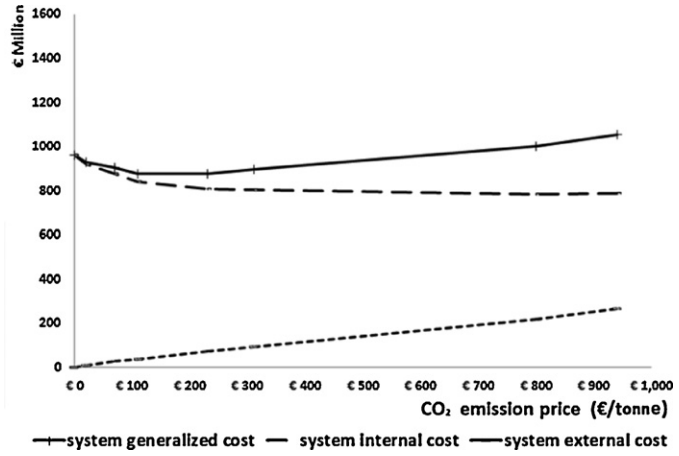


Fig. 7. System costs of optimized terminal network configurations under CO_2 emission prices ranging €0–€1000 per tonne.

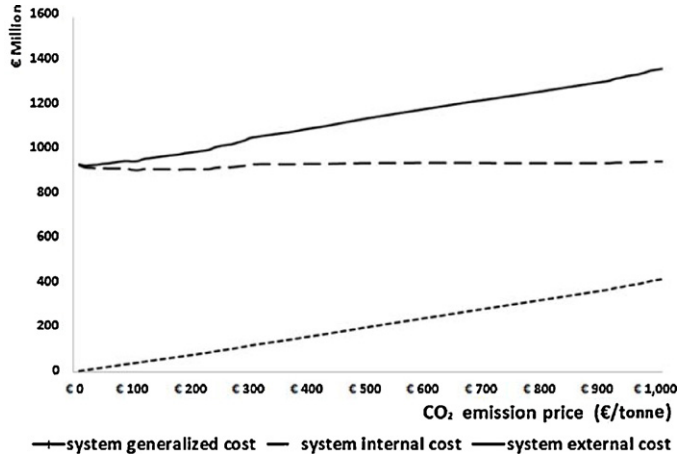


Fig. 8. System costs under CO_2 emission prices ranging €0–€1000 per tonne without the effects of economies of terminal scale.

3.3. Findings from network optimization

This section shows our main findings on terminal network optimization under various CO_2 emission prices, in terms of CO_2 emissions, system costs, and network configuration.

The relationship between system CO_2 emissions and the CO_2 emission price is shown in Fig. 6. An advantage of GA is that it provides a broader range of results instead of only a frontier. The dots are spread over in a large range on the axis of the system CO_2 emissions, which indicates a strong impact of the terminal network configuration on the system CO_2 emissions. The frontier marked in the figure, consisting of the terminal network configurations, shows the minimal system CO_2 emissions that are able to be achieved by pricing CO_2 at the certain rates. The frontier goes downwards with an increase in the CO_2 emission price. This indicates that CO_2 pricing is an effective policy in terms of reducing the CO_2 emissions in the condition of a reasonable terminal network configuration. Furthermore, the slope of the boundary suggests the efficiency of CO_2 pricing policy in terms of reducing the system CO_2 emissions.

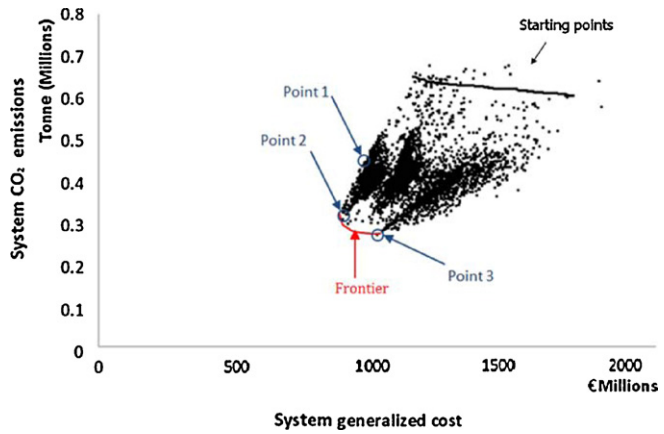


Fig. 9. System generalized costs and system CO_2 emissions of terminal network configurations.

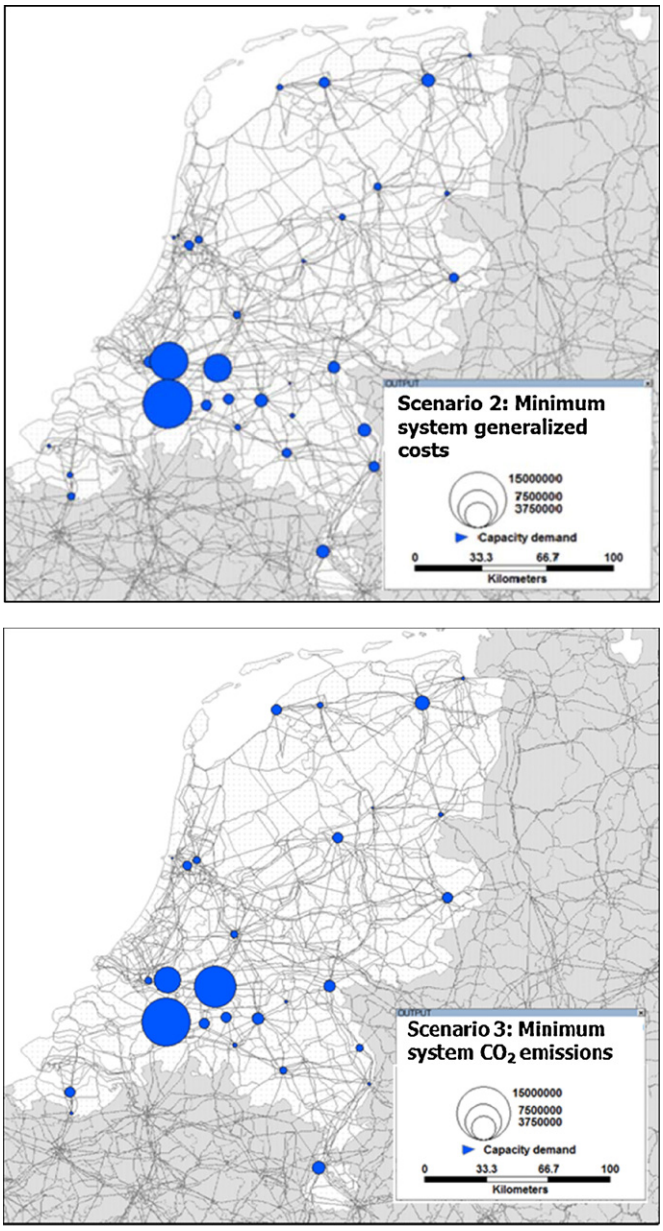


Fig. 10. Examples of terminal network configurations: (1) scenario of optimized current terminal configuration; (2) scenario of minimum system generalized costs and (3) scenario of minimum system CO_2 emissions.

Fig. 7 shows the trend of system generalized costs, system internal costs, and system external costs when the CO_2 emission price increases. Here, the internal costs are the generalized costs excluding the CO_2 emission costs, and the external costs, where only take into account the costs caused by CO_2 pricing.

The system generalized costs decrease at the beginning and reach the inflection point at between €100 and €150 per tonne of CO_2 emissions. This is owing to the combined effect of model shift and economies of terminal scale. If CO_2 emissions are more expensive than the price at the inflection point, the cost-saving gained from modal shift and economies of scale cannot compensate for the extra charges on CO_2 emissions and overall system costs increase again. A comparison with another test, where we assumed that the economies of scale do not exist in the system, confirmed that the economies of scale are the dominant factor leading to the decrease in the generalized costs (see Fig. 8). We conclude that the system costs can be reduced by appropriately introducing external costs into the system, if the internalization of the external costs would lead to terminal scale economies. To our knowledge this is a new finding that could have important implications for freight transport policy. Further tests of the model with respect to the conditions under which this phenomenon occurs are needed, however, before policy recommendations can be made.

Fig. 9 shows the generalized costs and CO_2 emissions of a large number of different terminal network configurations with different CO_2 emission prices.

The curve shown in the figure illustrates the frontier of the efficient solutions, consisting of optimal terminal network configurations under CO_2 emission prices ranging €0–€1000 per tonne. Each point in the figure represents a combination of a specific terminal network configuration and a specific CO_2 emission price. Three key points are highlighted. Point 1 indicates the current scenario where there are no extra costs on CO_2 emissions and all terminals are open; Point 2 shows the scenario of minimum system generalized costs; and Point 3 is the scenario of minimum system CO_2 emission. The terminal network configurations of the scenario of minimum system generalized costs and the scenario of minimum system emissions are shown in Fig. 10, in terms of number of terminals, locations, and the associated capacities. We found that the preferred terminal network configuration depends on the CO_2 emission price. In order to achieve different objectives, for example, minimum system generalized costs or minimum system emissions, different network configurations are required.

4. Conclusions

This paper has proposed a GIS-based model for the optimization of multimodal multicommodity freight terminal network systems, and taking into account terminal economies of scale and CO_2 pricing. The optimization problem is solved by bi-level programming, where the upper level searches for the optimal combination of terminal network configuration and CO_2 price in order to achieve minimal system cost, while the lower level performs multi-commodity flow assignment over a real-scaled multimodal network. GA-based heuristic algorithms were used in both the upper level optimization and the parameter estimation. The multimodal assignment model was estimated on terminal throughput and freight transport survey data and is able to reproduce observed flows.

The model is applied to optimize the Dutch container terminal network configuration in terms of minimizing the system generalized costs and minimizing the system CO_2 emissions under different CO_2 prices. The model shows the effect of changing CO_2 prices on terminal network configuration, generalized costs and system CO_2 emissions. Optimizing network configurations for different CO_2 prices results in a range of contrasting network solutions in terms of terminal locations and sizes. System costs can be reduced substantially by introducing external costs into the system.

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